

Analysis of the number of offences in Swiss cantons: economic and social variables associated with crime trends in 2021

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Executive Summary

Our project focuses on the impact of two socio-economic variables, GDP per canton and the proportion of individuals in an asylum situation, on the number of offences reported in Swiss cantons in 2021. We analyze the 26 cantons that make up the Swiss Confederation.

Criminal acts are not just indicators of crime; they also reflect deeper social dynamics, such as economic constraints, social disparities and demographic trends. Switzerland, with its 26 cantons, offers a unique opportunity to analyze local differences and identify possible links between crime and socio-economic variables.

We start from two assumptions, which are the following:

- > A higher GDP leads to a higher number of offences, as wealthier cantons may experience higher living costs and economic disparities likely to favor offences.
- > A higher proportion of asylum seekers is correlated with a higher number of offences, as asylum seekers face limited rights and difficult circumstances.

Using statistical tools such as correlation analysis, scatter plots and contingency tables...we evaluated these hypotheses.

The results revealed :

A strong positive correlation (0.988) between GDP and the number of offences, confirming the first hypothesis. Wealthier cantons tend to report more offences, potentially linked to greater economic activity and disparities.

A weak correlation (0.116) between the proportion of individuals in an asylum situation and the number of offences, indicating that there is no significant relationship. This calls into question the second hypothesis and suggests that asylum seekers are not a major factor influencing offence rates.

The number of offences in Swiss cantons can be influenced by certain socio-economic variables, such as GDP, but this is not always the case, as can be seen with the proportion of individuals in asylum situations. Consequently, no generalizations should be made, and further research is needed to better understand these dynamics.

1. Introduction

Offenses are not only an indicator of crime. Indeed, they reveal deeper trends that drive society, shedding light on its economic pressures, social inequalities and demographic aspects. Consequently, offenses are a relevant subject for a deeper analysis on reported rates in Swiss cantons in 2021. Despite the popular statistical approach of the issue, this subject needs a deeper approach to determinate trends and activities that have an influence on safety and social wellbeing.

Nevertheless, due to its federal structure with 26 cantons, Switzerland is an ideal setting to study local differences. Each canton, with its own characteristics, enables analyses of the links between different factors as well as between different parameters and the number of reported offenses.

Thus, we have asked ourselves, which factors can explain the variations of the number of offenses between Swiss cantons?

Consequently, in this study, the number of offenses is the main variable while the associated variables have been chosen for their relevance : the GDP per canton and the proportion of individuals in asylum situations:

Hypothesis 1: When the GDP is higher, the number of offenses increases. So, for cantons with a higher GDP, the cost of living would be higher. Individuals with lower wealth would be more likely to commit offenses.

Hypothesis 2: When the proportion of people in asylum situations is high, the number of offenses is higher. Individuals in asylum situations have fewer rights than other residents of the canton. As example, they don't have the right to work, which makes their already difficult situations even more so.

2. Data gathering and description

Firstly, the three variables are continuous quantitative variables.

_For the number of offenses per canton in 2021: This main variable represents the total of offenses which encompasses all types of crimes as defined by the Swiss penal code such as: homicides, thefts, money laundering... (See appendix A). We chose to measure the value in number of individuals instead of the proportion of individuals. For the proportion, we would have had to divide the value by the number of people living in the canton which doesn't consider the fact that offenses can be committed by people living outside of the canton or even Switzerland.

_For the GDP per canton in 2021: This concerns Gross Domestic Products and can be defined in multiple ways depending on how it is calculated. In this case, as per the OFS that provides these numbers, the GDP is defined as the sum of the added value that different products and services take (*Produit intérieur brut, n.d.*). The GDP is calculated to take inflation into

account, allowing comparison of its evolution from one period to another. It is measured in thousands of CHF. The GDP represents the economic operations of the cantons and its monetary value, which shows the wealth and the cost of living of said cantons. It will be interesting in our study to see how a canton's wealth is linked to the offense rate. Also, because this variable is expressed in proportion, it helps the comparison between canton.

Proportion of individuals in asylum situations (*Secrétariat d'État aux migrations SEM, n.d*) :

This variable measures the portion of the total population of the canton, that is in the process of obtaining asylum. This includes asylum seekers who have applied but whose case is still being processed. It is calculated by dividing the number of individuals in asylum situations by the number of residents in the canton. It is a proportional value expressed in decimals, for example: The values like 0,006619 indicates a proportion which means that 0,006619% of the population is concerned. It allows us to see if a correlation or a relation exists between the variable "Proportion of individuals in asylum situations". The proportion facilitates comparisons between cantons concerned by the variable.

3. Methodology

To analyse our data we used R software , intended for statistics .We are mainly spotting the relations between the number of offences and the previously introduced variables ; proportion of individuals in the asylum situation and the GDP per canton .

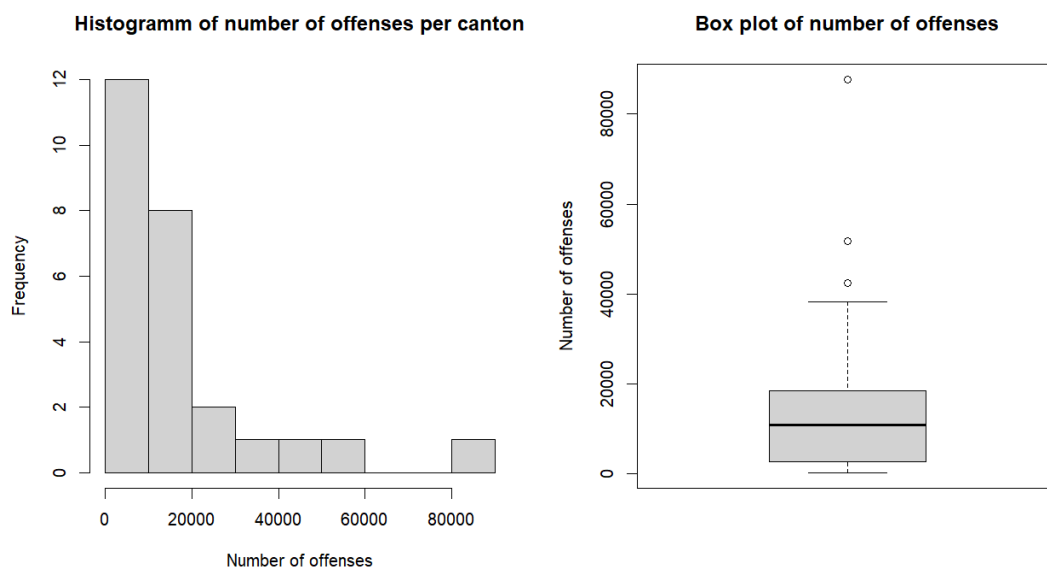
At first , we will study each variable in a more detailed way ; histograms ,box plot, average , median ,variance and extremes .

Then ,we will study the association between the number of offences and the two others variables .We will use for example scatterplots and joint tables .

4. Analysis

4.1 Analysis of the main variable

Figure 1 : Diagrams of the main variable



First, the histogram shows that the distribution of the number of offenses in our population has a positive asymmetry. This means that most cantons have a somewhat low number of offenses, less than 30 000 (22 cantons). The modal class corresponds to a number of offenses between 0 and 10 000 with 12 cantons in this class.

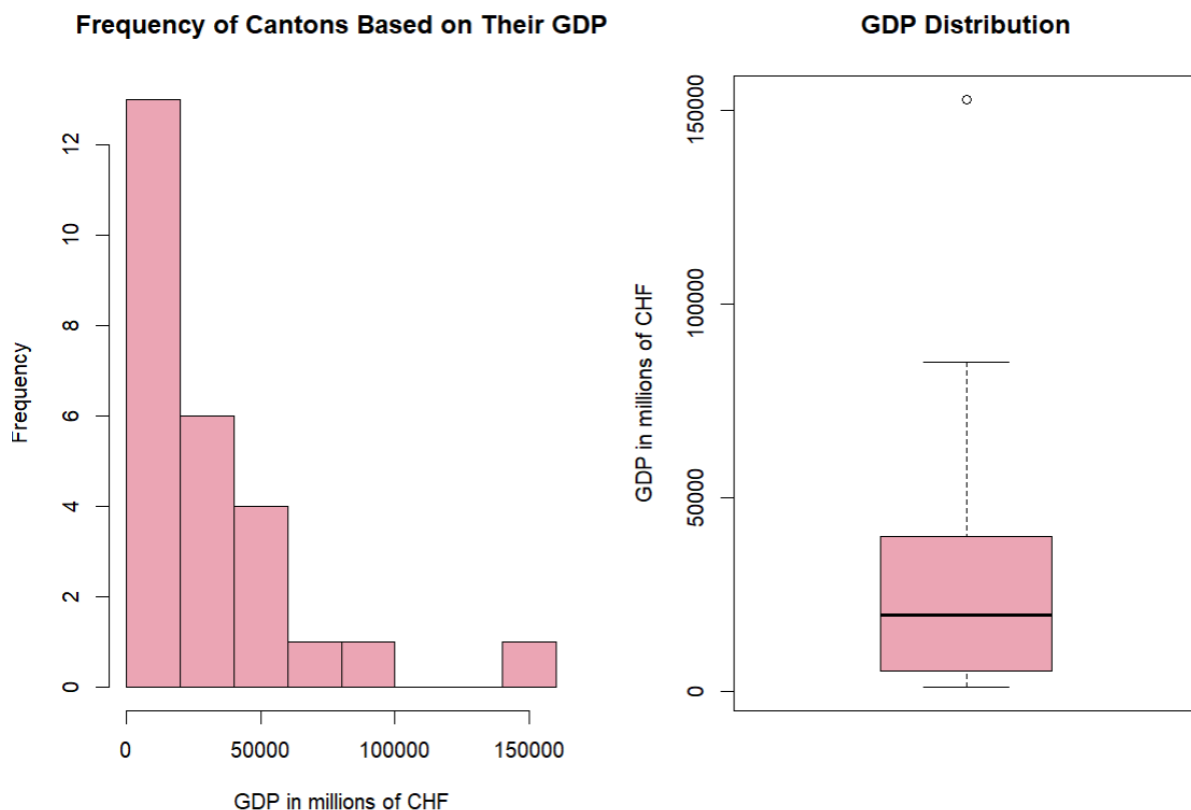
Next, the boxplot shows that there are three cantons with outlier values concerning the number of offenses. These outliers correspond, in ascending order, to the cantons of Vaud, Bern and Zurich. The box is quite shallow which means that 50% of the value is not widely distributed.

The mean for this variable is 15 942 ,5 offenses, but taking into account the outliers, we will use the median. The median is 10 890,5 offenses and the standard deviation is 19 930,58. That means on one hand, that generally Swiss cantons don't have a big number of offenses per year and on the other hand, the number of offenses per canton is quite homogenous since the standard deviation isn't too high. The 90th percentile is 40 367,00 and shows that only ten percent of the cantons exceed this value. (See the quantiles in appendix B).

In short, we realize that a lot of cantons don't have a big problem with criminal activities and are relatively safe. A minority of cantons are less safe because there are more offenses.

4.2 Presentation of the associated variables

Figure 2 : Diagram of the associated variable : the GDP



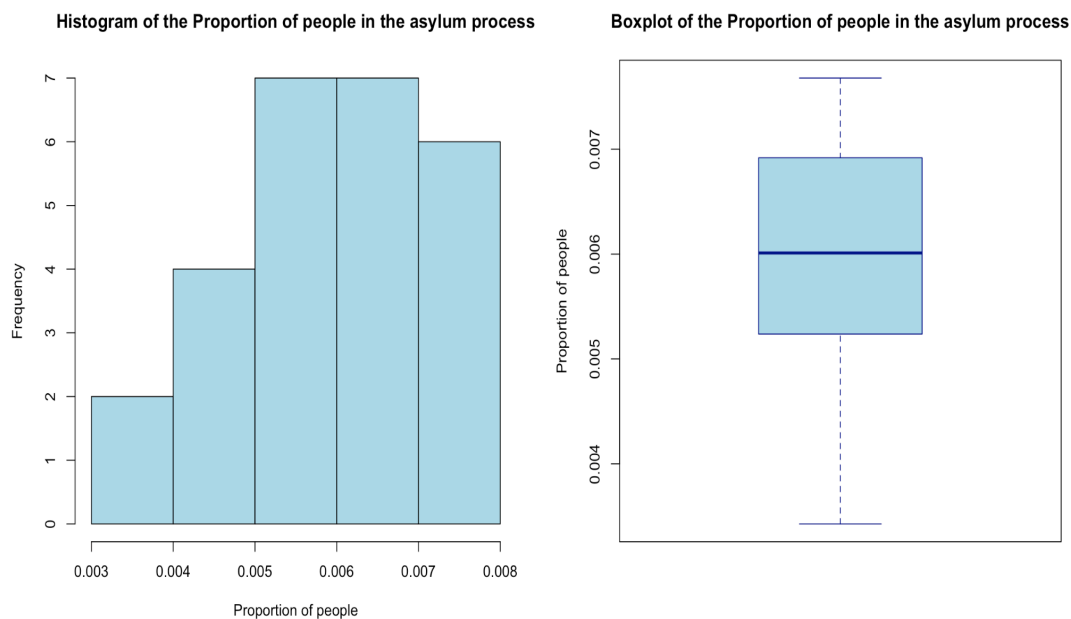
We can observe the positive asymmetry in this graph, that almost 19 cantons have a GDP under 40 000 mio of CHF and only one canton has a GDP higher than 100 000 mio of CHF. The modal class corresponds to a GDP between 0 and 20 000, with 13 cantons in this class. According to the boxplot, we can see an outlier which is the canton of Zurich with a value of 152 941 mio of CHF as GDP.

This value explains the difference between the mean value of 28 656,5 mio CHF and the median of 19 867 mio CHF.

According to the calculations, the variance is 1088454299 and the standard deviation is equal to 32 991,73. That means that the GDP per canton is quite homogenous since the standard deviation isn't too high (See the quantiles in appendix B).

In short, we can see that most cantons have a not too high GDP and are in a quite good economic situation and that only a minority of them have a very high GDP.

Figure 3 : Diagram of the associated variable : the proportion of people in the asylum process



The histogram reveals a slight negative asymmetry with a low frequency of cantons (2) for the very low proportions (0,003 to 0,004). However, we can observe that a majority of cantons have a proportion between 0,005 and 0,007 of people in the asylum process .

The boxplot allows us to see that there is no outlier value. The 25th percentile has a value less than 0,005052. The median is less or equal to 0,006015.

The mean of the proportion is 0,0059943446, indicating generally the cantons have a very low proportion of people in asylum situations compared to the total population.

The variance is 1.44e-06 which means the standard deviation is 0.0012 . This means that the proportion of people in asylum situations is quite dispersed.

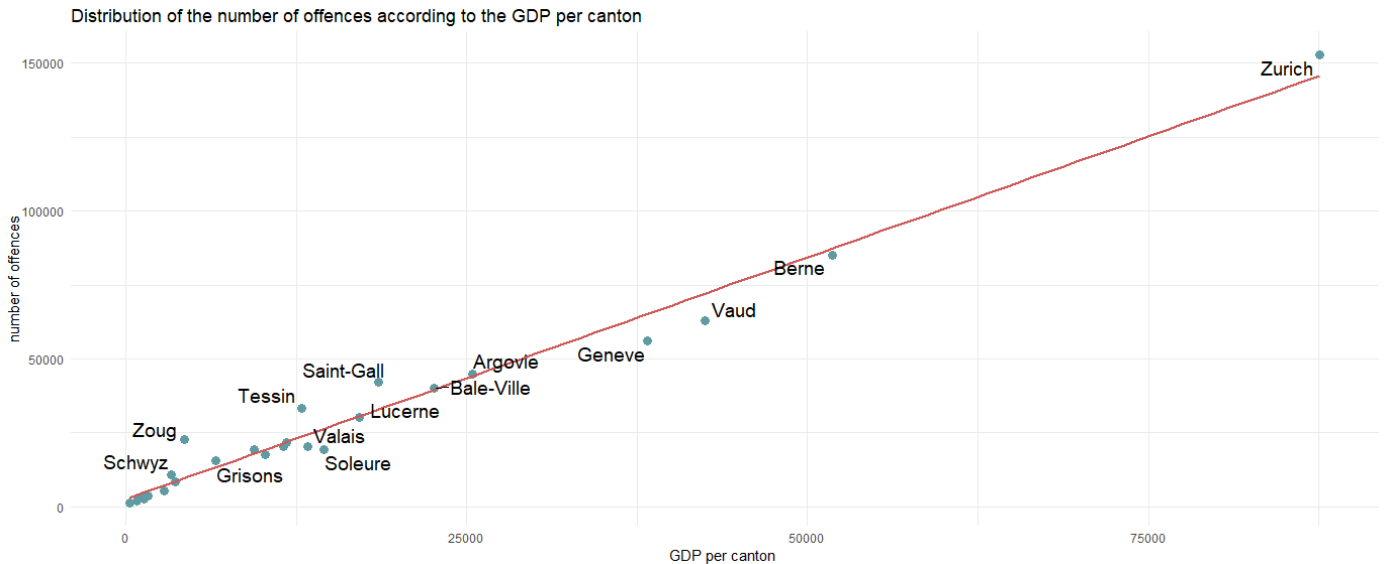
In short, the proportion of asylum seekers in the cantons is generally low and evenly distributed between the cantons.

4.3 Research of a potential association between the main variable and the associated variables

4.3.1 Link between the number of offences and the GDP per canton

Figure 4 : Diagram of the relation between the numbers of offences and the GDP

(Alboukadel.,2019)



After visualizing the scatter plot of the number of offenses and the GDP for each canton, we can notice that generally, as the GDP grows higher, the number of offenses tends to be greater. So, we have a positive correlation between the two variables which is around 0,988 (as calculated in the R file). This rather big value shows a strong relation between the two variables as we can see in the graph below: the red line, which corresponds to the linear regression using the least square method, approximates the scatter plot well.

Table 1 : Joint contingency table for the variable “Number of offenses” and “GDP in millions of CHF” (See R code)

X the variable for the number of offenses and Y the variable for the GDP in million CHF

X	Y	Y < 50000 (low GDP)	50000 < Y < 100000 (medium GDP)	Y > 100000 (high GDP)	Total
X ≤ 10000 (low number of offenses)		8	4	0	12
10000 < X ≤ 40000 (medium number of offenses)		0	7	4	11
X > 40000 (high number of offenses)		0	0	3	3
Total		8	11	7	26

The contingency table shows that the majority of cantons (8) have a low GDP and a low number of offenses. There are no cantons with a low GDP who have a high number of offenses. Also 7 cantons have an average GDP and an average number of offenses. The cantons with a high GDP have a number of offenses that are average to high and are overall a minority.

Research of the dependency between the main variable and the GDP per canton

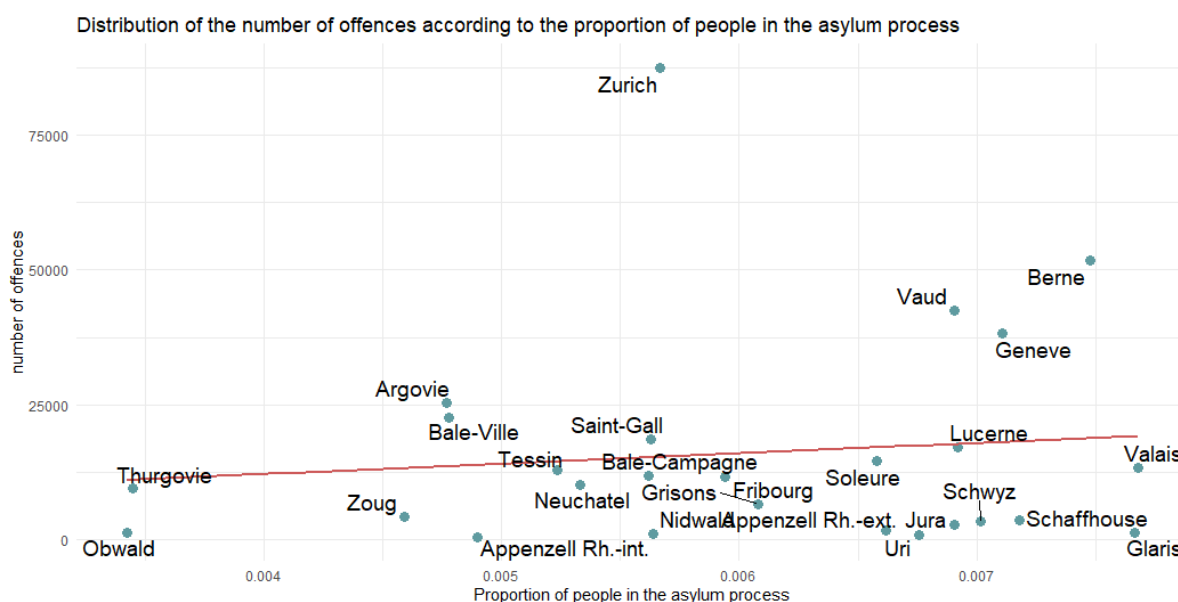
Base on the jointed density table (See annex C) we know that the probability for a canton to have a medium number of offenses is 0,423

For the probability to have a canton to have a medium number of offenses knowing that the GDP is high is : $P(10000 < X \leq 40000 | 50000 < Y < 100000) = P(10000 < X \leq 40000 \& 50000 < Y < 100000) / P(50000 < Y < 100000) = 0,636$.

Since 0,423 is different from 0,636, we can conclude there is a dependency between the variables.

4.3.2 Link between the number of offenses and the proportion of individuals in asylum

Figure 5 : Diagram of the relation between the numbers of offences and the proportion of people in the asylum



Concerning the analysis of the number of offenses and the proportion of individuals in asylum situations. The correlation between these two variables is around 0,116 which is close to 0. We therefore have little relation between those two variables. Indeed, as we can see on the graph below which is a scatter plot between those variables, the dots don't seem to follow a curve but are rather dispersed.

This graph also shows that there are a lot of offences in the canton of Zürich, but that the proportion of people in the asylum process in this canton is rather close to the average.

Table 2 : Joint contingency table for the variable “Number of offenses” and “the proportion of individuals in asylum” (See R code)

X	Y	Y2<=0.005	0.005<Y2<=0.007	0.007<Y2	Total
X <=10000 (faible nombre d'infraction)		4	5	3	12
10000< X <=40000 (nombre d'infraction moyen)		2	7	2	11
X > 40000 (nombre d'infraction élevé)		0	2	1	3
Total		6	14	6	26

The contingency table shows the distribution of the offenses (X) according to the proportion of individuals in asylum (Y2).

It shows that the canton with the lowest proportion of people in asylum, got either a low level or a medium level of offenses. Then, the majority of cantons, 7 cantons, got a proportion of individuals in asylum between 0.005 and 0.007 and a medium number of infractions. The second biggest part in this table is for the same interval for the Y2 variable but with a low number of offenses, 5 cantons. Then finally for the highest interval for the Y2 variable there are a few number of cantons in the column. It is with the minimum number of offenses that we have more offenses in this column.

To sum up, we notice that, it is not necessarily that with a higher proportion of individuals in asylum that we got a higher number of offenses.

Research of the dependency between the main variable and the proportion of individuals in asylum”

Based on the jointed density table (See annex C) we know that the probability for a canton to have a relatively high number of offenses is 0,115.

For the probability to have a canton to have a high number of offenses knowing that the Y2 is relatively medium is : $P(X > 40000 | 0,005 < Y2 < 0,007) = 0,143$.

Since 0,115 is different from 0,143, we can conclude there is a dependency between the variables.

4.4 Limitations

_The main variable evaluates the number of offenses as defined by the Swiss penal code. According to the results, it is impossible to determine if a specific type of offense is more linked to our variable than another. This means that our results can't tell on which offenses we would need to act upon to have more impact.

_After the analysis of the different variables, we noticed that the individuals in asylum situations are not strongly linked to our main variable. This comes from the fact that their proportions are very low. For example, in canton Bern, they are 0,7% of total inhabitants.

_The data used were measured during 2021, the year marked by covid-19. This can limit the relevancy of our results, as during this time, the socioeconomic trends indicators like the number of offenses and the GDP were altered. Our results may not be valid in a more economically stable context.

_For the variable “Individuals in asylum situations”, in the original database, there were 1696 people that were not assigned a canton and that were therefore not analyzed.

_Since the variable for the number of offenses is not in proportion, it is a bit difficult to compare effectively this variable between cantons.

5. Discussion

Through our analysis, we explored the relationships between socio-economic variables and the number of offences reported in Swiss cantons. For example, we found that a higher GDP is associated with a higher number of offences, which seems to confirm our first hypothesis. This high correlation (0,988) could be explained by increased economic activity or greater wealth disparities within the richer cantons.

We also examined the link between the proportion of individuals in an asylum situation and the number of offences. Here, the weak correlation (0,116) observed suggests that there is little to no relation. This finding calls into question our second hypothesis and indicates that asylum seekers are not a significant factor influencing crime rates, contrary to some common perceptions based on their difficult situation. But the result is also because there are very few of them in the cantons.

The second hypothesis is therefore more difficult to confirm. Although we have identified certain trends, the lack of solid evidence points to the need for further research to better understand the dynamics at play.

Our analysis has highlighted some significant links, notably between GDP and crime rates. So the cantons should study at a deeper level their inequality in income distribution because a high GDP seems to mean that more people have hard times providing basic necessities. Contrary to the proportion of the people in the asylum process that cantons should not worry about in this case.

However, it also revealed that socio-economic variables do not always have clear or direct impacts on crime patterns. This underscores the importance of nuanced research to better understand these relationships and develop appropriate policies.

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7. Appendix – Supplements

A . Différentes catégories et sous catégories de délits

Offenses Related to Life and Bodily Integrity: Consummated homicides (Art. 111–113/116), Attempted homicides (Art. 111–113/116), Serious bodily harm (Art. 122), Simple bodily harm (Art. 123).

Offenses Related to Property: Theft (excluding vehicle theft) (Art. 139), including burglary, snatching theft, Vehicle theft (Art. 139), Robbery (Art. 140), Property damage (excluding theft) (Art. 144), Fraud (Art. 146), Extortion and blackmail (Art. 156), Offenses in bankruptcy and debt collection (Art. 163–171).

Offenses Related to Honor and Privacy: Defamation and slander (Art. 173 + 174).

Offenses Related to Freedom: Threats (Art. 180), Coercion (Art. 181), Human trafficking (Art. 182), Kidnapping and abduction (Art. 183), Trespassing without theft (Art. 186).

Offenses Related to Sexual Integrity: Sexual acts with children (Art. 187), Rape (Art. 190), Exhibitionism (Art. 194), Pornography (Art. 197).

Offenses Related to Collective Danger: Intentional arson (Art. 221).

Offenses Related to Public Authority: Violence or threats against authorities or officials (Art. 285).

Offenses Related to the Administration of Justice: Money laundering (Art. 305bis).

Other unmentioned offenses were also considered.

B . Tables for the quartiles

Table 3 : Quartiles for the main variable

	10%	25%	50%	75%	90%
Valeurs	1126,00	2909,75	10890,50	18175,00	40367,00

Table 4 : Quartiles for the variable of the the GDP in mio CHF

	10%	25%	50%	75%	90%
Valeurs	2862.00	6090.25	19867,5	38360.75	59634.00

Table 5 : Quartiles for the variable of the proportion of individuals in asylum situation

	10%	25%	50%	75%	90%
Valeurs	0.00468000	0.00526050	0.00601050	0.00691475	0.00732650

C . Jointed contingency table**Table 6 : Joint density table for the variable “Number of offenses” and “GDP in millions of CHF” (See R code)**

X	Y1	Y1 < 50000	50000< Y1 < 100000	Y1 > 100000	Total
X <=10000 (low number of offenses)		0,308	0,154	0,000	0,462
10000< X <=40000 (medium number of offenses)		0,000	0,269	0,154	0,423
X > 40000 (high number of offenses)		0,000	0,000	0,115	0,115
Total		0,308	0,423	0,269	1

Table 7 : Joint density table for the variable “Number of offenses” and “Proportion of the individual in asylum” (See R code)

	Y2<=0.005	0.005<Y2<=0.007	0.007<Y2	Total
X <=10000 (low number of offenses)	0.154	0.192	0.115	0.462
10000< X <=40000 (medium number of offenses)	0.077	0.269	0.077	0.423
X > 40000 (high number of offenses)	0.000	0.077	0.038	0.115
Total	0.231	0.538	0.231	1

8. Appendice – Code R

Code R pour l'analyse de la variable principale

```
#histogram and boxplot
par(mfrow = c(1, 2))
hist(Data$number.of.offences,
      main="Histogram of number of offenses per canton",
      xlab="Number of offenses",
      ylab = "Frequency",breaks=7)
boxplot(Data$number.of.offences,main="Box plot of number of offenses",
        ylab = "Number of offenses")

#average,median and standard deviation calculation
mean(Data$number.of.offences)
median(Data$number.of.offences)
sd(Data$number.of.offences)

#Calculation of quantiles
quantile(Data$number.of.offences, prob = c(0.1, 0.25, 0.5, 0.75, 0.9))

#Sorting the number of infractions in descending order
sort(Data$number.of.offences, decreasing = T)[0:3]

# finding the 3 cantons corresponding to the highest number of infractions
subset(Data, number.of.offences == 87586, , select = "Cantons")
subset(Data, number.of.offences == 51813, , select = "Cantons")
subset(Data, number.of.offences == 42500, , select = "Cantons")
```

CODE R, FOR THE ANALYSIS OF THE SECOND ASSOCIATED VARIABLE : The proportion of person in asylum process

```
par(mfrow = c(1, 2))

#histogram and boxplot

hist(Data$proportion.of.people.in.the.asylum.process,
      main = "Histogram of the Proportion of people in the asylum process",
      xlab = "Proportion of people",
      ylab = "Frequency",
      col = "lightblue",
      border = "black")

boxplot(Data$proportion.of.people.in.the.asylum.process,
        main = "Boxplot of the Proportion of people in the asylum process",
        ylab = "Proportion of people",
        col = "lightblue",
        border = "darkblue")

#average,median and variance calculation

mean(Data$proportion.of.people.in.the.asylum.process)

median(Data$proportion.of.people.in.the.asylum.process)

var(Data$proportion.of.people.in.the.asylum.process)

#Calculation of quantiles

quantile(Data$proportion.of.people.in.the.asylum.process, prob = c(0.1, 0.25, 0.5, 0.75, 0.9))
```

R CODE FOR ANALYZING THE THIRD ASSOCIATED VARIABLE : GDP in millions of CHF by canton.

```
par(mfrow = c(1,2)) #divide the screen in two parts

#histogram and boxplot
hist(Data$PIB,
      main="Frequency of Cantons Based on Their GDP",
      xlab="GDP in millions of CHF",
      ylab = "Frequency",
      col='pink2')
boxplot(Data$PIB,main="GDP Distribution",
        ylab = "GDP in millions of CHF",
        col='pink2')

#average,median ,variance and standard deviation calculation
x=Data$PIB
mean(x)
var(x)
sd(x)

#Calculation of quantiles
quantile(x, prob = c(0.1,0.25, 0.5, 0.75,0.9))
```

Joint contingency and joint density table for the 2 associated variables:

```
head(Data)
X = Data[, "number.of.offences"]
Y = Data[, "PIB"]
Z = Data[, "proportion.of.people.in.the.asylum.process"]

X1 = X
X1[X<= 10000] = 1
X1[10000 < X & X <= 40000] = 2
X1[X> 40000] = 3

Y1 = Y
Y1[Y <= 10000] = 1
Y1[10000 < Y & Y<= 40000] = 2
Y1[Y > 40000] = 3
Z1 = Z
```

```

Z1[Z <= 0.005] = 1
Z1[0.005 < Z & Z<= 0.007] = 2
Z1[Z > 0.007] = 3

#contingency table for the number of infraction and the GDP
contingencejointeY = matrix(0, 3, 3)
for (i in 1:3) for (j in 1:3) {
  contingencejointeY[i, j] = sum(X1 == i & Y1 == j)}
colnames(contingencejointeY) = c("PIB<=50000", "50000<PIB<=100000", "100000<PIB")
rownames(contingencejointeY) = c("NB infrac<=50000", "50000<Nb infrac<=100000", "100000<Nb
infrac")
contingencejointeY

#density table for the number of infraction and the GDP
densitejointeY = matrix(0, 3, 3)
for (i in 1:3) for (j in 1:3) {
  densitejointeY[i, j] = sum(X1 == i & Y1 == j)/26
}
colnames(densitejointeY) = c("PIB<=50000", "50000<PIB<=100000", "100000<PIB")
rownames(densitejointeY) = c("NB infrac<=50000", "50000<Nb infrac<=100000", "100000<Nb
infrac")
densitejointeY

#density table for the number of infraction and the proportion of people in the asylum process
densitejointeZ = matrix(0, 3, 3)
for (i in 1:3) for (j in 1:3) {
  densitejointeZ[i, j] = sum(X1 == i & Z1 == j)/26}

colnames(densitejointeZ) = c("Prop of people<=0.005", "0.005<Prop of people<=0.007",
"0.007<Prop of people")
rownames(densitejointeZ) = c("NB infrac<=10000", "10000<Nb infrac<=40000", "40000<Nb infrac")
densitejointeZ

#contingency table for the number of infraction and the proportion of people in the asylum
process
contingencejointeZ = matrix(0, 3, 3)
for (i in 1:3) for (j in 1:3) {
  contingencejointeZ[i, j] = sum(X1 == i & Z1 == j)}

colnames(contingencejointeZ) = c("Prop of people<=0.005", "0.005<Prop of people<=0.007",
"0.007<Prop of people")
rownames(contingencejointeZ) = c("NB infrac<=10000", "10000<Nb infrac<=40000", "40000<Nb
infrac")

contingencejointeZ

```

Scatter plot for the correlation of the variables

```

#installation des packages nécessaires pour le nuage de points annoté
install.packages("ggplot2")
install.packages("ggrepel")
library("ggplot2")

```

```
library("ggrepel")

# linear regression between the proportion of people in the asylum process and the number of
offences.
regression = lm(Data$number.of.offences
~Data$proportion.of.people.in.the.asylum.process,data=Data)

#scatter plot annotated with the names of the cantons.
ggplot(Data, aes(x =proportion.of.people.in.the.asylum.process, y =number.of.offences, label
=Cantons )) +
  geom_point(color = "cadetblue", size = 3) +
  geom_smooth(method = "lm", color = "indianred", se = FALSE) +
  geom_text_repel(size = 5) +
  theme_minimal() +
  labs(title = "Distribution of the number of offences according to the proportion of people in the
asylum process ", x = "Proportion of people in the asylum process", y = "number of offences")
```

```
#word cloud between number of offences and PIB

# linear regression between GDP and the number of offences
regression = lm(Data$number.of.offences ~
Data$PIB,data=Data)

#scatter plot annotated with the names of the cantons.
ggplot(Data, aes(x =number.of.offences,
y =PIB, label =Cantons )) +
  geom_point(color = "cadetblue", size = 3) +
  geom_smooth(method = "lm", color = "indianred", se = FALSE) + # add regression
labs(title = "Linear regression: Frequency vs. length",
x = "Word length",
y = "Frequency") + # Add points
geom_text_repel(size = 5) + # Add labels (avoids overlapping)
theme_minimal() + # Graphic style
labs(title = "Distribution of the number of offences according to the GDP per
canton ", x = "GDP per canton ", y = "number of offences")
```

Déclaration d'honneur

Par la présente, j'affirme avoir pris connaissance des documents d'information et de prévention du plagiat émis par l'Université de Neuchâtel et m'être renseigné-e correctement sur les techniques de citation. J'atteste par ailleurs que le travail rendu est le fruit de ma réflexion personnelle et a été rédigé de manière autonome.

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Neuchâtel, Décembre 2024

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